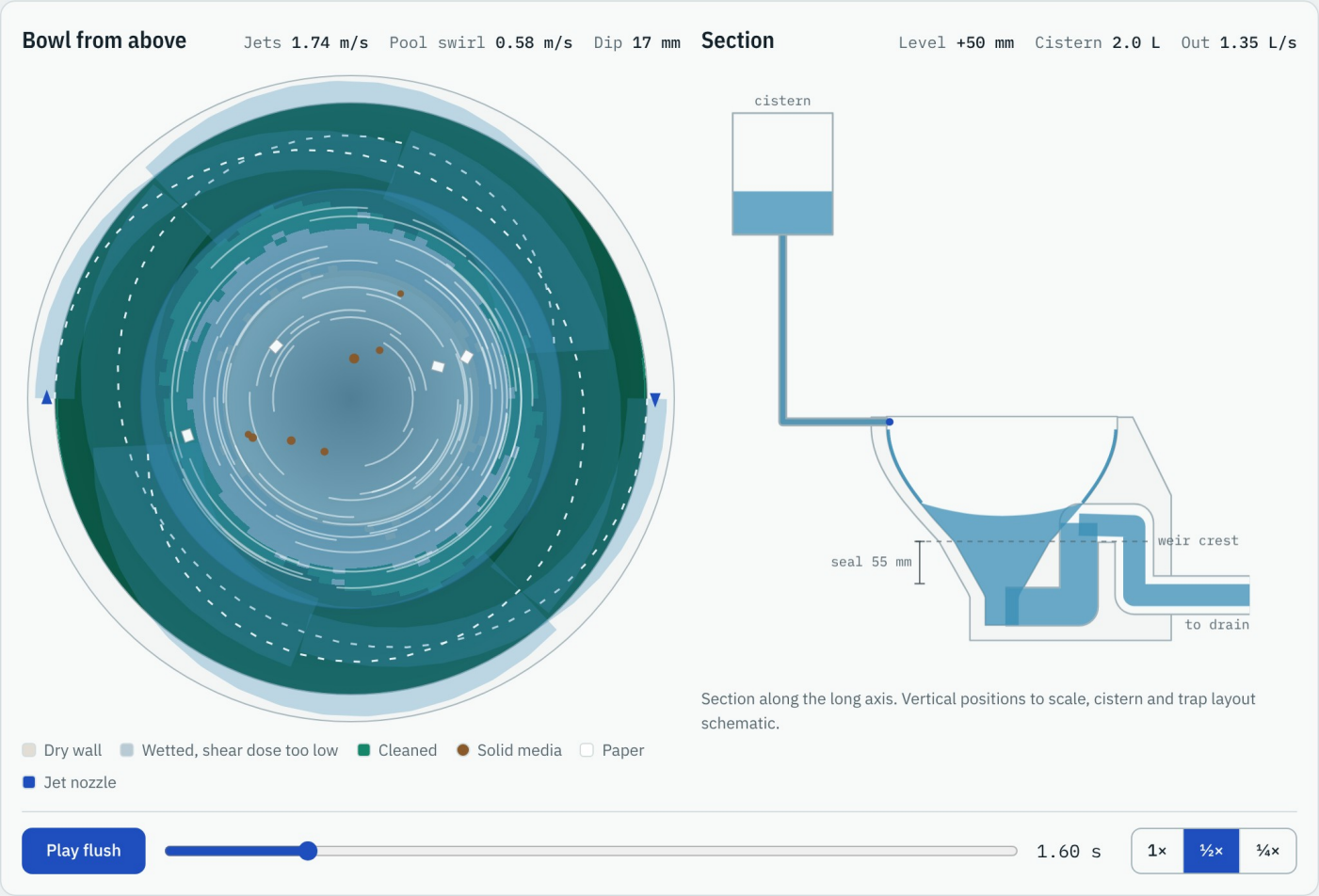
Tangential jet flush model

One flush computed in milliseconds instead of hours of free-surface CFD



88 %

of the bowl wall above the waterline cleaned

99 %

of a 250 g solid test load removed

4.5 L

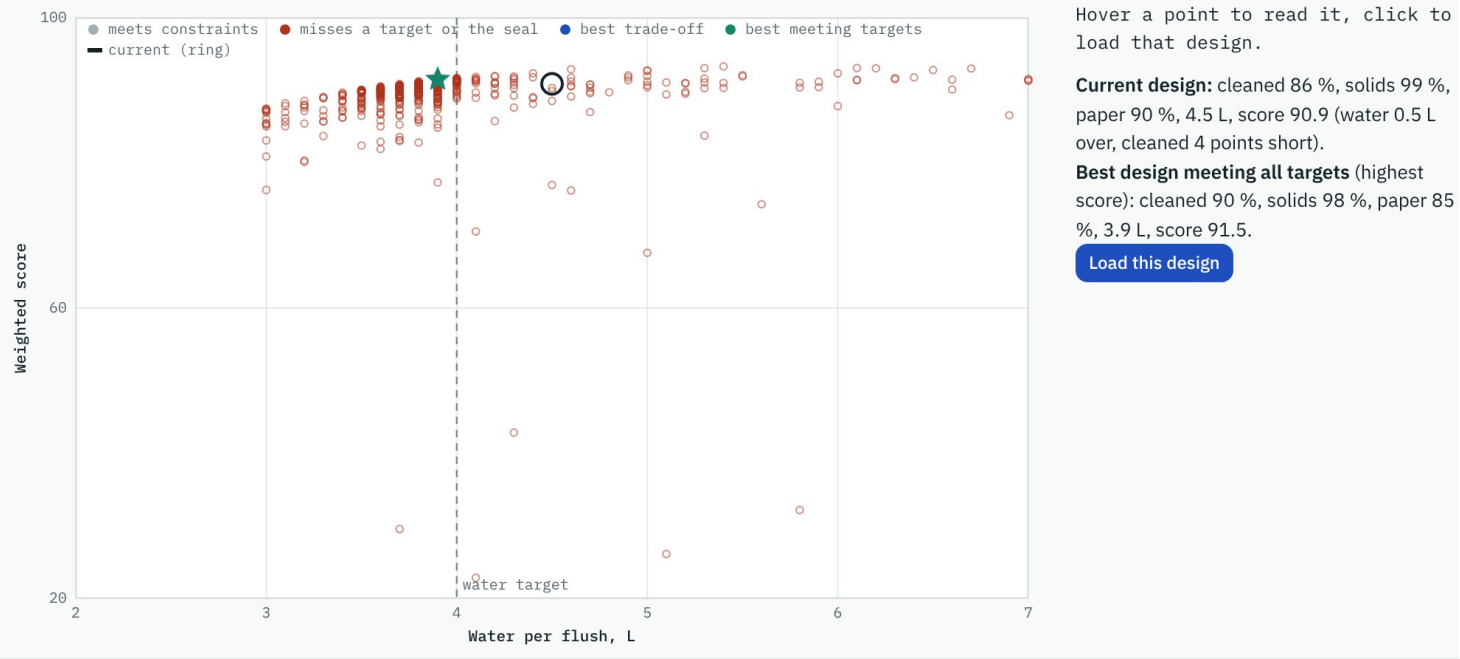
water per flush, EU washdown, full flush

Try it live
fintlp.github.io/jet-flush-model

Reduced-order model. Coefficients are not yet calibrated against CFD or rig data, so results show trends between designs, not validated values.

AI lab: let the model search

Three ways AI works with the physics model, all running in the browser



Each dot is one simulated flush; red rings miss a target. The blue curve is the best trade-off between score and water, and the green star is the best design that meets all targets.

5–20 ms
physics model, per flush

≈ 5 µs
neural surrogate, per design

minutes to hours
PreonLab or CFD, per case

1 Design optimiser

Finds the best settings you allow, or a design that meets targets such as at most 4.0 L and at least 90 % cleaned. About 540 flushes in 4 seconds.

2 Neural surrogate

Learns the model from 300 flushes and answers about a thousand times faster. In practice it would be trained on PreonLab runs.

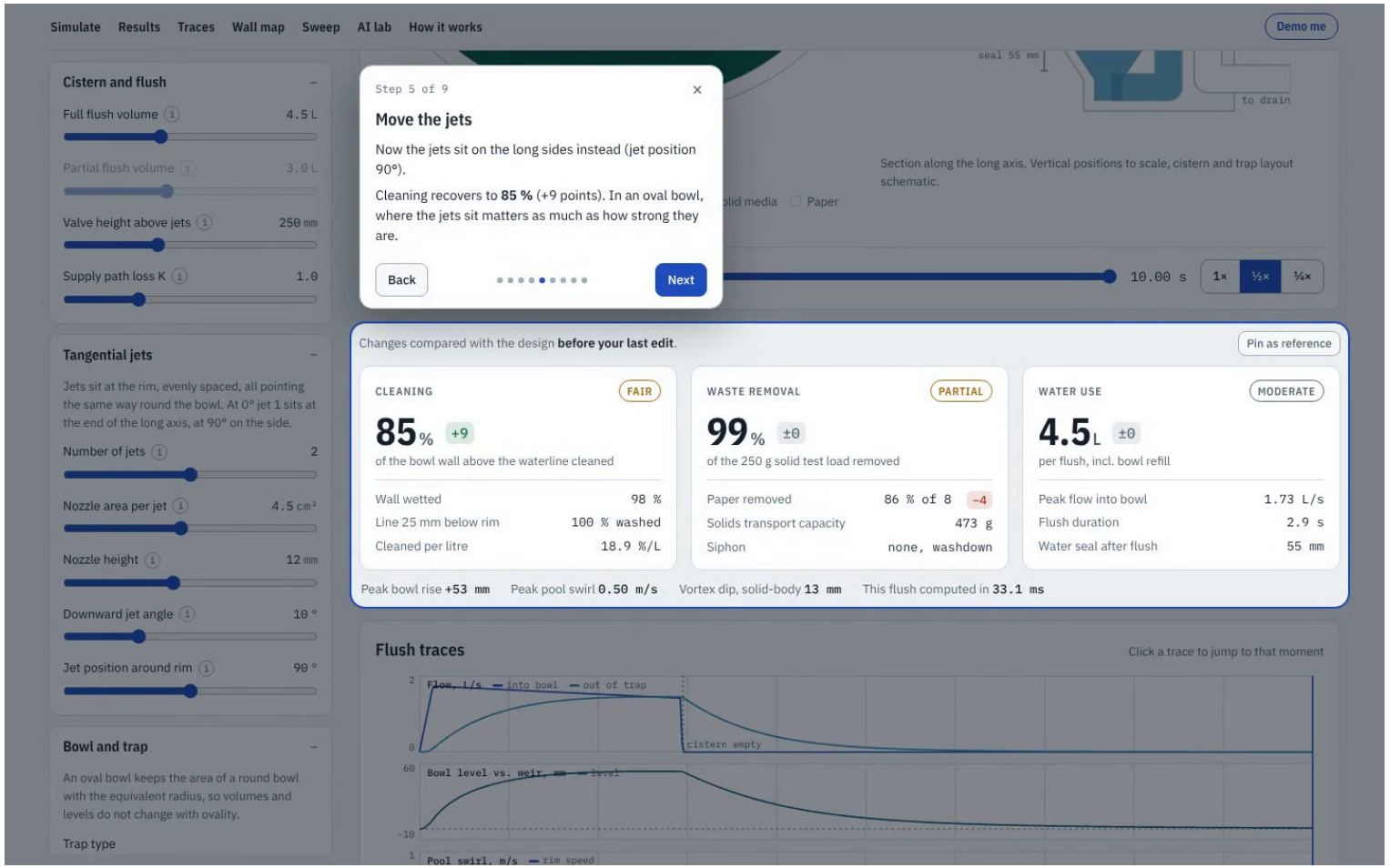
3 Hybrid calibration

Fits the model to a few reference runs and corrects the rest with data. Cuts the error on unseen designs by about two to eight times.

Results come from the uncalibrated model: suggestions are candidates for a CFD or rig check, not final answers. The calibration demo uses synthetic reference data.

Guided demo in one minute

Play a flush, make the bowl oval, let the AI find a better design, all live in the browser



88 % → 76 %

cleaning when the bowl turns oval, jets at the ends

76 % → 85 %

after moving the jets to the long sides

90 % at 4.0 L

AI lab design: more cleaning, 0.5 L less water

Start it yourself
fintlp.github.io/jet-flush-model/#demo

Recorded from the live page on 19 October 2026. Numbers come from the uncalibrated model.